

Xiaoshan Wu

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EDUCATION

- The University of Hong Kong (HKU)** Hong Kong
Ph.D. in Electrical and Electronic Engineering (Focus: 3D/4D Vision, World Models & Embodied AI) Sep. 2023 – Present
- **Awards:** Hong Kong PhD Fellowship (**HKPFS, Top Tier**), HKU Presidential Scholarship.
 - **Advisor:** Dr. Xiaojuan Qi. **GPA:** 4.0/4.3.
- Zhejiang University (ZJU) & UIUC** China & USA
Dual B.S. in Computer Engineering (ZJUI Institute) Sep. 2019 – Jun. 2023
- **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

Self-Improving Physical Agents through Agent–World Model Co-Evolution. I connect robust perception, action-conditioned 4D world models, and scene/experience memory to turn limited physical interaction into cumulative, transferable capability, enabling agents to choose informative experiments, evaluate persistent consequences, and reuse verified successes, failures, and recoveries.

PUBLICATIONS & PREPRINTS

* indicates equal contribution; † denotes project lead.

1. **X. Wu**, R. Cai, Y. Bao, Z. Su, A. Wu, F. Wu, Z. Li, Z. Qi, H. Chen, W. Ye, J. Zhang, K. Zhou, M. Wang, and Y. Du. **Verifiable 4D World–Action Model for Robust Robot Control.**
Under review at the International Conference on Learning Representations (ICLR), 2027.
2. **X. Wu**, X. Lyu, Y. Yu, M. Liu, S. Shi, Z. Wang, and X. Qi. **GaMe: Geometry as Memory for Multi-View Spatial Reasoning.**
Under review at the IEEE International Conference on Robotics and Automation (ICRA), 2027.
3. Y. Yu*, **X. Wu***†, X. Lyu, Z. Wang, and X. Qi. **EVSplat: Uncertainty-Aware Gaussian Splatting for Dynamic Scenes with Event Camera.**
Under review at the AAAI Conference on Artificial Intelligence, 2027.
4. T. Wan*, **X. Wu***†, Y. Yu, et al. **LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.**
arXiv:2609.25803, 2026. Under review at IEEE TPAMI.
5. **X. Wu**, X. Lyu, Y. Yu, B. Wang, Z. Wang, and X. Qi. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.**
International Conference on Learning Representations (ICLR), 2026.
6. Y. Yu, **X. Wu***, X. Hu, et al. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.**
arXiv:2512.04519, 2025.
7. **X. Wu**, Y. Yu, X. Lyu, et al. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.**
NeurIPS, 2025 (Spotlight).
8. **X. Wu**, W. He, M. Yao, et al. **Event-Based Depth Prediction With Deep Spiking Neural Network.**
IEEE Transactions on Cognitive and Developmental Systems, vol. 16, no. 6, pp. 2008–2018, 2024.
9. M. Liu, X. Lyu, T. Ren, et al. **FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.**
European Conference on Computer Vision (ECCV), 2026.
10. M. Liu, T. Liu, J. Xia, et al. **OptiGeo: Efficient Monocular Geometry for Embodied Perception in Optically Challenging Scenes.**
arXiv:2608.29881, 2026.
11. X. Lyu, M. Liu, **X. Wu**, et al. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7577–7587, 2026.
12. Y. Yu, X. Zhang, S. Wang, et al. **Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.**
Nature, vol. 654, pp. 642–651, 2026.
13. B. Wang, X. Zhang, S. Wang, et al. **Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.**
Science Advances, vol. 11, no. 16, eads5340, 2025.
14. H. Chen, J. Yang, J. Chen, et al. **Continuous-Time Digital Twin with Analog Memristive Neural Ordinary Differential Equation Solver.**
Science Advances, vol. 11, no. 22, eadr7571, 2025.
15. C. Li, X. Chen, Z. Zhang, **X. Wu**, et al. **Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.**
Nano Letters, vol. 24, no. 47, pp. 15025–15034, 2024.

RESEARCH EXPERIENCE

- Harvard University (SEAS)** Cambridge, MA, USA
Visiting Researcher Apr. 2026 – Aug. 2026
- Collaborated with Yilun Du and Mengyu Wang on self-improving physical agents and agent–world model co-evolution.

TECHNICAL SKILLS

Research Expertise: Self-Improving Physical Agents, World–Action Models, 3D/4D Vision, Spatial Reasoning, Video Generation, Event Vision.
Frameworks & Tools: PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS)	2023
HKU Presidential Scholarship	2023
Outstanding Graduate of Zhejiang University	2023